

Unit: DAM
2024-2025

DAM Project *"E-commerce shopping App"*

Build an e-commerce mobile application using Flutter where users can search, filter and browse stores and their products. The app will simulate an online store experience with a clean, modern UI and basic backend integration.

Students are free to choose which types of stores/products to use on the app.

Actors:

Admins – Each admin manages a store, on which he can add, remove and delete products.

Clients – Each client can view a specific store, and then searches/filters products based on specific filters and categories. Products details (price, description, colors, sizes...etc) are shown upon clicking on a product, which then can be added to a cart (that can be viewed at any time). He can also view and modify his profile, and make comments/ratings/likes on each product.

Some Hints:

- Flutter framework must be used.
- You can use SQL database with flutter directly.
- Note that you can rely on your own imagination, you do not have to follow these sprints.
- You may use **any type of database** you want (firebase, NoSQL, SQL...etc. **except for SQLite**),

As a reminder, **REST API SQL** Connection and usage will be presented as a course chapter.

- a functional DB login page, as well as how to add and manage items on a cart will be done as exercises on TP5.
- Note that Sprints 1, 2 and 3 are mandatory to get the minimum mark. Doing sprints from 4 to 5 will increase the obtained mark obviously.

The project will be marked as 11/20 of the total TP mark. For more transparency, each sprint has its own respective mark. Two(2) Marks are to be awarded for the UI design, creativity, and other non-mentioned used APIs and functions (examples : notifications, stripe API...etc.)

Sprint 1: Stores and Products Display App (2 points)

- Design a basic, user-friendly interface.
- Implement a store selection tool with visual icons or images.
- Display Products within each selected store, divided into several categories.
- Allow users to view a list of products under each category.
- Design simple detail pages for each product with a photo and information (name, price, description, size...etc.).

Sprint 2: Admin Panel & Content Management(1.5 points)

- Design a user-friendly admin interface (with login if you wish).
- Allow admins to add, edit, or delete products and their categories for their respective stores.

Sprint 3: Product details (2 points)

- Allow users to rate products on a scale (e.g., 1 to 5 stars).
- Display average ratings on each products' page
- Implement a "like" or "bookmark" feature for users to save their favorite products.
- Design a basic review system where users can leave comments about products alongside their ratings.
- Display user comments, number of likes and average rating on each products' detail page.

Sprint 4: Backend(2.5 points)

- Register users in the database and develop a login/signup process.

- Save user information (email, password, favorite products...etc.), products and stores (their images, likes, comments, and feedbacks), and all their data on the database.
- Fetch and modify data from the database when using the app.
- Note that you do not have to fully do all tables; you may do two tables for instance and get marked partially for them.

Sprint 5: Enhanced Search & Filtering Capabilities(1 point)

- Develop a search bar with predictive text and suggestions.
- Implement filters to help users narrow down results based on categories, products type or number of likes/comment of a product.

The project is to be returned on May, exact date will be sent later but probably on the first week. Feel free to email me your concerns if you have any.

Each project should contain no more than Three (3) students; all **students should be present during the project's consultation.**

Students can be from different groups.